

Post-Maneuver UCT Correlation Using Multi-Source Data Streams

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ABSTRACT

Katalyst Space Technologies has identified efficient and autonomous uncorrelated track (UCT) resolution workflows to be a key bottleneck in the ability to quickly identify large and potentially consequential maneuvers of spacecraft. A UCT can be resolved when it is re-associated to a previously known object track and the full state history of the object becomes available. However, current automated systems falter when they encounter maneuvers that break the kinematic constraints of observation correlation algorithms, leading to UCTs. Traditionally, observation association algorithms rely upon purely kinematic and non-probabilistic correlation algorithms, which propagate the orbital dynamics of a catalog of objects and compare the propagated states with observations. These algorithms set thresholds around the propagated orbit, and generally only consider the position and velocity of objects, leaving out additional observed information such as visual magnitude. This approach can fail when correlating across a maneuver because, based on state alone, once the deviation exceeds the thresholds, it often fails to correlate with any objects in the catalog. This causes UCTs to drop into manual correlation processes executed by analysts, resulting in hours of delays trying to resolve them.

To overcome these challenges, Katalyst has explored a post-maneuver UCT resolution pipeline that integrates both kinematic and non-kinematic correlation techniques to resolve UCTs with all available information collected by Electro-Optical (EO) sensors. The workflow incorporates statistical distributions of observable quantities generated from historical observations on objects in the catalog, which can be compared to the corresponding values in the UCT observations to improve the probability of correlations. The Post-Maneuver UCT resolution architecture is composed of five main components. First, Stochastic Initial Orbit Determination (SIOD) creates a probability distribution for the state of the object based on the UCT observations. Second, Candidate Determination generates UCT-object candidate pairs, filtering based on thresholds derived from the SIOD step. Then, Probabilistic Kinematic Correlation determines the feasibility of correlation for a candidate pair using forward and backward propagation and track correlation. In parallel with previous steps, Statistical Object Identification compares the visual magnitude and solar phase angle of each observation of the UCT to prior distributions of cataloged objects. Lastly, Confidence Determination computes the correlation confidence for each candidate pair using a machine learning model that considers the kinematic feasibility and the scores from Statistical Object Identification. In conclusion, this workflow can correctly identify previously correlated tracks simulated as UCTs with small maneuvers with a precision of 95% and a recall of 97%. The performance with the baseline dataset demonstrates the baseline feasibility of the proposed probabilistic UCT correlation workflow, and demonstrates the utility of incorporating non-kinematic information to improve the confidence in correlations.

1. INTRODUCTION

A critical requirement for achieving robust space domain awareness (SDA) is the ability to process and reduce the number of uncorrelated tracks or UCTs (a set of observations of a trackable object whose origin has not yet been established), by making associations to a known catalog of objects.

Traditionally, when an unknown object maneuvers and changes its orbit, this complicates tracking and degrades ephemeris accuracy. Maneuvers, along with launches and break-up events, are the main source of new object detections during Resident Space Object (RSO) cataloging activities [1]. Tracking and detecting maneuvering objects makes correlating observations to known objects more difficult. UCT correlation is usually performed by comparing real orbits from cataloged RSOs and predicted UCT orbits, using propagation of states to identify possible candidate

object matches that minimize error. This method requires a relatively long observation history because the correlation process is likely to fail for the first track after a sufficiently large maneuver or after enough time passes to significantly increase the errors of the predicted orbit. Because of this data requirement for traditional UCT resolution, the time to re-associate an object after it is lost is often delayed.

If a track cannot be associated to a previously known object, the full state history is lost, and a maneuver cannot be detected or characterized. This leads to ambiguous correlations or a loss of custody for RSOs, creating UCTs or duplicated RSOs in the catalog. Once a track fails to correlate it becomes a UCT, and is passed to a process is usually done manually by human analysts examining orbits to find potential intersections and identify a maneuver. While analysts possess the skills to correlate UCTs and detect maneuvers, their bandwidth constraints and the time required to develop these insights clearly illustrates the urgent requirement for tools that autonomously process new observations and optimize processing to provide the rapid maneuver detection that manual processes cannot.

Currently, there are thousands of operational satellites in orbit and thousands more planned. Most of these satellites perform maneuvers every one or two weeks but, in some cases, up to four times a day. As the frequency of satellite maneuvers increases along with the amount of catalog comparisons, track custody maintenance will become more difficult, leading to growing uncertainty in the whereabouts of spacecraft and an increased risk of collisions. Unidentified maneuvers impact everyone operating in the domain including commercial and government assets that rely on the Department of Commerce for Space Traffic Management (STM). Research by COMSPOC has shown there is degradation in the accuracy of orbital tracking caused by maneuvers, including the risks from unannounced rendezvous and proximity operations — such as those being routinely performed by the Russian “inspector” satellite Luch/Olymp [2]. Timely detection of RSO movements and assessment of their risk to satellite assets is imperative for STM. The rising number of aggressive activities and dangerous maneuvers by satellites underscores this need to provide timely and accurate SDA intelligence [1]. Automated characterization and downstream processing will be foundational for hazard mitigation in the space domain.

2. BACKGROUND

Research on UCT resolution and maneuver detection and characterization spans a breadth of approaches. Pastor et al. developed a sequential filtering algorithm to identify new objects from a simulated scenario using optical measurement data to associate UCTs belonging to the same object [3]. These methods have previously been used for radar tracks, but optical track association has different challenges due to the reduced orbital information obtained from a single track. Instead of using a brute force approach by evaluating all possible combinations of UCTs, several filters and complexity reduction techniques are typically used to reduce the computational resources required. Building on this research, Pastor et al. developed two novel methods for the detection and estimation of impulsive maneuvers in GEO [4]. The first method performs track-to-orbit (T2O) associations to detect and estimate single burn maneuvers by determining the pre-maneuver orbit that minimizes the post-maneuver observations residuals. The second method addresses the orbit-to-orbit (O2O) association problem by approximating maneuvers of two burns as linear perturbations over the nominal ballistic motion of RSOs. These methods could be applied to optical or radar sensors, and passive ranging stations. A single track is not enough for the T2O methodology to reliably estimate a maneuver so the tracks must be associated before applying the T2O algorithm.

Multi Target Tracking (MTT) methods have traditionally been used for sensing, guidance, navigation, and air traffic control. The detection of maneuvers can be framed as a multi-target tracking-association problem. Kelecý et al. and Lemmens et al. ingest information from Two Line Element (TLE) data to isolate, and thus detect, maneuver intervals [5, 6]. Jia et al. used an interacting multiple model (IMM) filter with different process noise levels to distinguish the maneuvering effects [7]. Specifically, an IMM cubature Kalman filter (IMM-CKF); which considers the geometric relations between the space object, space based optical (SBO) sensor, and the Sun; was used to track a maneuvering space object. Their results indicate that IMM-CKF is more robust than the CKF when the space object undergoes a maneuver. In addition, the detection of a maneuver can be obtained by using the IMM-CKF. These methods were only tested with a single scenario with four SBO sensors. Additionally, Singh et al. used an optimal control framework as a post-processor to a multiple hypothesis tracker for resolving UCTs generated by maneuvers [8]. The approach is most applicable to routine, but unannounced fuel-optimal maneuvers. Satellite operators may not always perform optimal maneuvers and therefore optimal control metrics should not be blindly trusted.

ROTAS (Report/Observation Association) is the standard application for maneuver detection within the Air Force Space Command (AFSPC) Astrodynamics Standard Software. ROTAS calculates the residuals for each observation against the associated element set to determine whether a maneuver has occurred [9]. Additionally, some existing companies that provide maneuver detection capabilities are Slingshot Aerospace, LeoLabs, COMSPOC, and ExoAnalytic Solutions. Slingshot developed a machine learning-based object profiling engine, which processes data from their global sensor network to monitor GEO spacecraft behavior. They then task their sensor network to validate the behavior identified [10]. LeoLabs has developed methods to detect and characterize maneuvers with lower and higher latency modes that can modify the detection window to identify maneuvers for high value objects or passively monitor a large catalog of objects. They demonstrated these methods on LeoLabs radar data [11]. ExoAnalytic has developed ExoALERT, an automated alerting system, designed and developed using data collected on GEO objects since 2012. Their service produces ephemeris data based on tracking data collected by their global telescope network which considers any known planned maneuvers [12]. COMSPOC has maneuver event detection and reconstruction techniques that use pre- and post-burn ephemerides [13]. The maneuver detection or characterization capabilities provided by these companies are either regime specific and/or specific to their sensor network.

Our research effort presented here includes the development of a post-maneuver UCT resolution workflow to resolve UCTs as soon as they are observed by prioritizing candidate comparisons using kinematic quantities as well as non-kinematic quantities from multiple electro-optical (EO) sensors. This paper explores a simulated architecture with a baseline dataset of correlated observations and small maneuvers to validate the methods for operational use on EO observations, element sets and state vectors. This workflow will be incorporated in our Advanced RSO Characterization (ARC) software for operational use with real-time data.

3. METHODS

3.1 Overview

Our primary research goal is to evaluate initial performance and establish baseline feasibility for our proposed UCT resolution pipeline, using correlated tracks as a stand-in for UCTs. A secondary research goal is to investigate the impact of including non-kinematic observable quantities, such as visual magnitude, as part of the confidence determination process for matching UCTs to known objects. We hypothesize that these non-kinematic quantities (which are generally reported along with positional coordinates, but not traditionally used in candidate matching) will provide a complementary approach to existing kinematic correlation methods. Thus, we predict an increase in the overall fraction of successful UCT matches for a combined approach over a kinematic-only correlation process for the same observations.

Here, we test our hypothesis by developing a UCT processing pipeline that incorporates both kinematic and non-kinematic correlation techniques. In Section 4, we compare this pipeline with a baseline that only includes the kinematic correlation procedures.

The five primary components of our hybrid UCT correlation pipeline are (1) Stochastic Initial Orbit Determination (SIOD), which generates an orbital probability distribution for a UCT from its positional observations; (2) Candidate Filtering, which narrows the region of parameter space to search for UCT-known object pairs based on the results from SIOD; (3) Probabilistic Kinematic Correlation, a determination of the relative likelihood that each filtered UCT-object pair is the best match by propagating their probability density functions (PDFs) to find their correlation probability at intermediate times; (4) Statistical Object Identification, a comparison of non-kinematic observed quantities of the UCT (e.g., visual magnitude and solar phase angle) with corresponding historical distributions across the RSO population; and (5) Confidence Determination, which computes the correlation confidence of each candidate pair using a Machine Learning (ML) model which takes the results of the previous steps into account (only the kinematic results for the baseline comparison).

Fig. 1 gives a schematic view of our UCT correlation pipeline. UCT data are ingested and preprocessed to assure they are properly formatted for evaluation by our pipeline. The SIOD, Candidate Filtering, and Probabilistic Kinematic Correlation steps are kinematic and the Statistical Object Identification is non-kinematic. Confidence Determination utilizes results from both kinematic and non-kinematic steps. UCT-object candidate pairs with confidences above a threshold value are flagged by our pipeline as a correlated pair, whereas, UCTs with less certain results will be given probabilistic correlation values for the most closely associated objects. We provide a detailed description of each step in our workflow in the following section (Section 3.2).

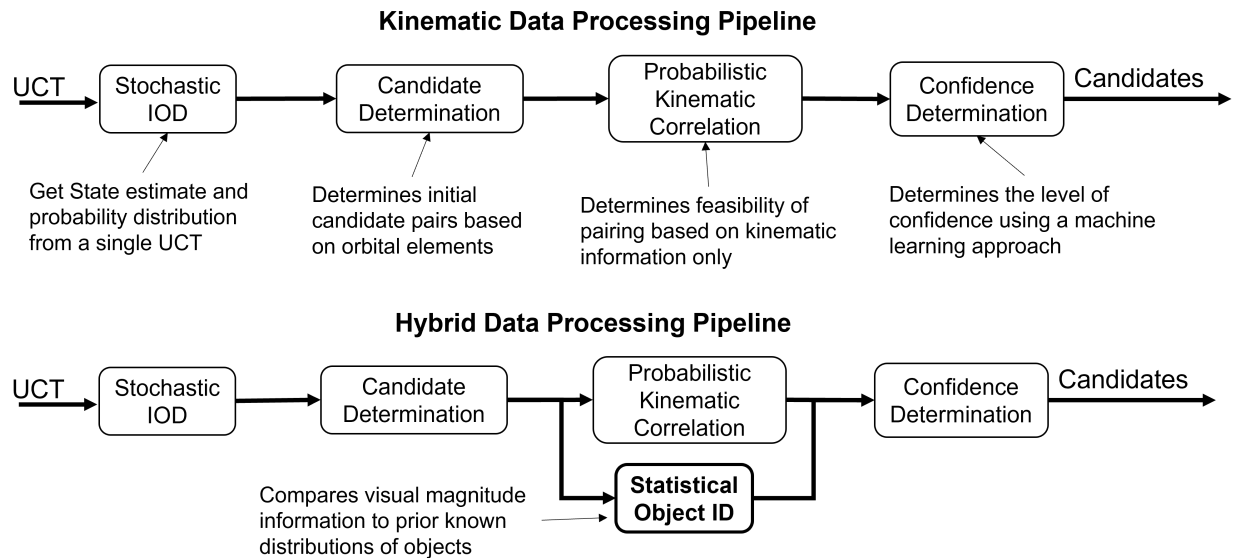


Fig. 1: The architecture of the kinematic and hybrid data processing pipelines. The hybrid pipeline illustrates the addition of visual magnitude and phase angle information that the Statistical Object Identification step contributes to confidence determination.

3.2 Data Processing Pipeline

To validate the effectiveness of our pipeline and to test our hypothesis, we curated a dataset of 2.1 million observations from the Unified Data Library (UDL) spanning the period from January 1 2024 to July 1 2024. These EO observations were sourced from ExoAnalytic Solutions (EXO) and InTrack Radar Technologies (IRT) Pine Park Observatory (PPEC). We also obtained 565,000 orbital element sets from the UDL, sourced from EXO, Cloudstone, and the 18th Space Defense Squadron (SDS).

We utilized the first four months of these data (January 1 2024 to April 30 2024) to train our statistical models (Section 3.2.5). Then, we used the final two months of data (May 1 2024 - July 1 2024) for evaluating our hypothesis. Splitting the data into two distinct time frames (a training set and a test set), rather than random sampling, provides a reliable estimate for future performance and is more representative of future operational systems (see, for example, [14]).

3.2.1 Preprocessing

Prior to running our pipeline, we preprocess the data to ensure it is correctly formatted. First, EO observations are split into “tracks” by dividing them into groups with the same object (satNo), observation site (idSensor), and no longer than 30 seconds between observations. For consistency, we disregard the TrackID field as it was not reported for all data sources. Instead, we created a trackNo field that assigns a unique number for each new sequence of observations when the time difference exceeds 30 seconds. We also applied a minimum track length of three observations and further filtered to only include correlated EO observations, i.e., observations with an associated element set.

In order to properly validate our results, we further limit the data to only correlated EO tracks that have associated elements sets within 24 hours prior to their observation. If multiple element sets are available within this timeframe, we use only the most recent one. The short timeframe prevents orbital propagation errors from dominating the analysis.

This level of correlation in our dataset is necessary to simulate UCTs for effectively testing our hypothesis. UCTs are, by definition, uncorrelated and, hence, cannot be validated except through additional observations which are beyond the scope of this work. So, ensuring our tracks are correlated, but processing them as if they were UCTs, allows us to quantify the effectiveness of our methods.

After these preprocessing steps, we perform a join across the database for all tracks and all element sets resulting in a set of “candidate pairs.” These candidate pairs represent every potentially feasible combination of track and element set

within the preprocessed data. For, each track there exists one corresponding element set that is the correct match. Each track was part of thousands of candidate pairs that are not correct matches. All candidate pairs can then be processed by our pipeline to determine their match probability. This allows us to validate our pipeline and quantitatively compare the kinematic-only baseline simulation results with the hybrid version that also includes non-kinematic quantities.

3.2.2 Stochastic Initial Orbit Determination

Once candidate pairs have been determined, our next objective is to fit an orbit to each individual track. In order to define an orbit for a given track, we need a set of six observable parameters, i.e., right ascension (RA), declination (Dec), range, and their corresponding rates. These parameters can be transformed into a Cartesian state vector and, from there, into a set of orbital elements applicable to the given track.

Typically, from ground based electro-optical observations, only two of the six necessary parameters are available and included in each track, the RA and Dec. We compute their rates by fitting a cubic spline to each as a function of time across the track and then calculating their derivatives at the mean time of the track.

Range and range-rate, are less straightforward to obtain and we estimate their values using the Constrained Admissible Region (CAR). The Admissible Region [15] is the area in range, range-rate space that encompasses orbits bound to the Earth (Fig. 2a). Within this region, we can place constraints on the range, semi-major axis, and eccentricity of the orbit in order to represent the known distribution of objects orbiting the Earth. We estimated this distribution using a sample dataset of orbital elements as in [16]. Each of these constraints corresponds to only allowing for range, range-rate value pairs within a portion of the Admissible Region. The resulting CAR then represents all plausible range, range-rate pairs for the orbit of the track.

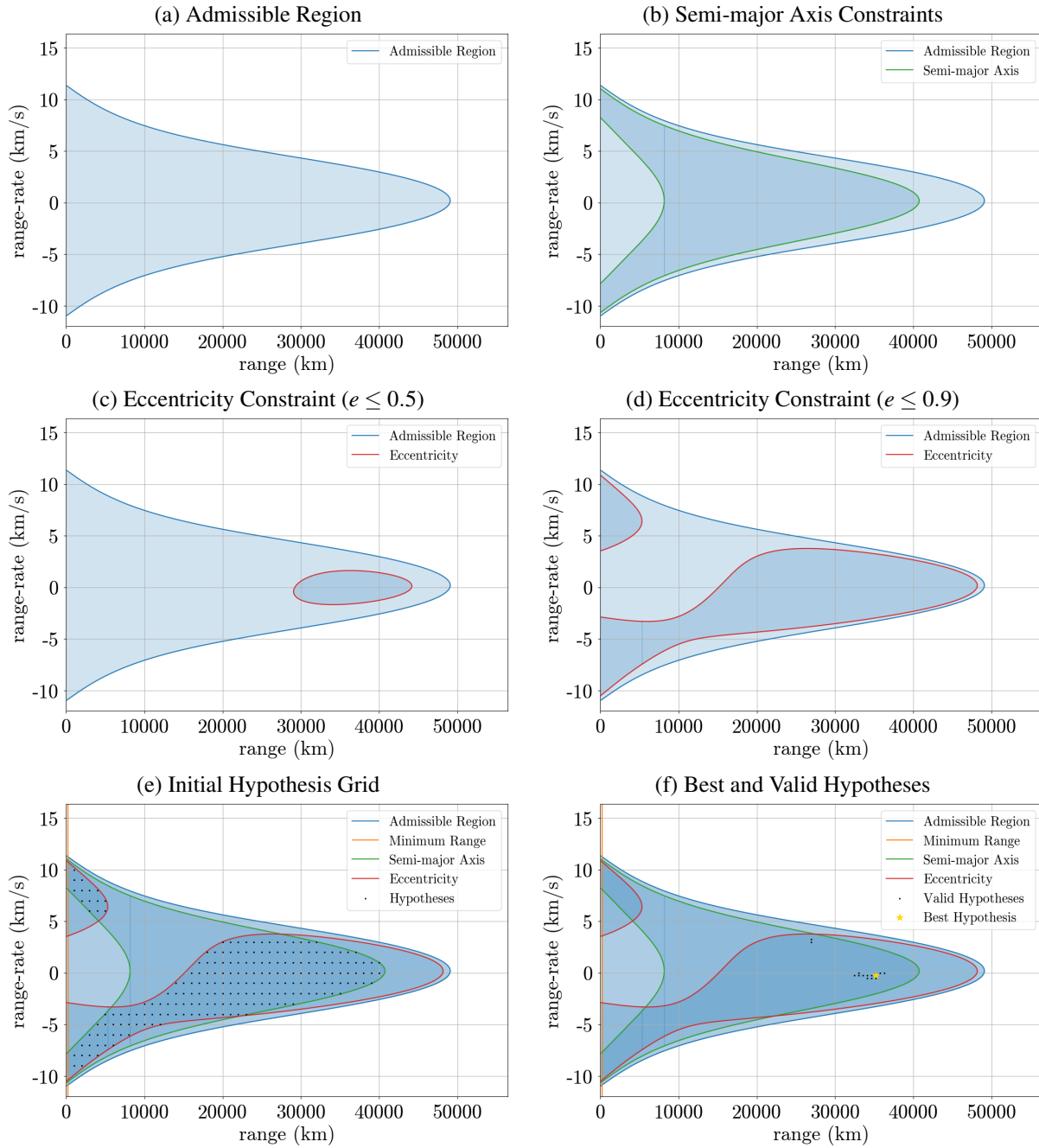


Fig. 2: Example constraints and hypotheses. (a) Bounding curves for the Admissible Region for objects orbiting the Earth. (b) Constraints on the maximum (right) and minimum (left) semi-major axis for the object. (c) A moderate eccentricity constraint with only two real roots. (d) A high eccentricity constraint with four real roots. (e) An initial hypothesis grid within the CAR. (f) Best/“Valid” hypotheses (black points) and the best-fit orbit (gold star) representing the minimum residual of the refined hypotheses.

Boundary curves for the CAR are computed as in [17, 18]. To constrain the semi-major axis and eccentricity, we solve for the range-rate as a function of range for a given value of the specific orbital energy. For the semi-major axis, this results in a set of bounding curves that encompass values between an upper and lower bound semi-major axis (Fig. 2b). The eccentricity constraint is also a function of the specific angular momentum which depends on both the range and range-rate, resulting in a quartic equation with either two or four real roots [17]. By pairing these roots, we

obtain bounding curves for either one or two sub-regions of the Admissible Region constrained by an upper bound on the eccentricity (Fig. 2c & d).

Once the CAR is defined, we generate a grid of hypotheses (range, range-rate pairs) within the bounds of the CAR (Fig. 2e). Each hypothesis represents a complete set of six parameters at the mean time of the track. From this parameter set, we convert to a Cartesian state vector (see, for example, [18]) and then to orbital elements (e.g., [19]).

Each hypothesis has a corresponding orbit fit, which we propagate to the times of each observation within the track using a Python implementation¹ of the SGP4 integrator [20]. By incorporating the observing site location coordinates, we can then determine a predicted RA and Dec at each observation time within our track for each hypothesized orbit. The difference between the measured and predicted RA and Dec are the residuals for each hypothesis.

Next, we evaluate each hypothesis using a loss function that calculates the sum of the squared Ra and Dec residuals. For each of the top-10 hypotheses with the lowest loss (minimum residuals), we draw 1,000 Monte Carlo samples in range, range-rate space from a Gaussian distribution centered on the hypothesis (see, for example [21]). We then select the best of these 1,000 points for each of the ten hypotheses and use those as the starting point for another Monte Carlo sampling, but with half the standard deviation of the first sampling. We repeat this process a total of three times and then select the best of these 10 “refined hypotheses” as our initial best-fit.

Once we have an initial value for the range and range-rate, we further refine this estimate by means of a gradient descent algorithm which directly optimizes the scaled Cartesian state (position and velocity, each normalized within the initial best-fit) to minimize the loss function. We then repeat this process 100 times, with stochastic variations in the RA and Dec based on their reported uncertainties, in order to build up a probability distribution representing our orbit fit. All of these orbital clones are propagated in the Probabilistic Kinematic Correlation step (Section 3.2.4) to generate the mean states and covariance matrices at intermediate timesteps.

To evaluate baseline performance of stages downstream in the pipeline, we generated a dataset of “synthetic SIOD” estimates. We generated these for each track in our dataset by propagating the most recent element set for the object to an intermediate time, adding a small randomized velocity offset to represent a maneuver, and then propagating the resulting state forward to the time of the track. At this point, we sampled 100 random particles from a representative 6-dimensional Gaussian distribution around this point. We then shifted the mean of the resulting particle cloud by a randomized offset drawn from a distribution with one tenth the standard deviation of the particle cloud. This representative distribution is informed by the results of running SIOD on real tracks. These “synthetic SIOD” results were used for baseline validation of downstream steps and training ML models, and a smaller dataset of “real SIOD results” were used for evaluation.

This SIOD process takes roughly 10 seconds per clone (1000 seconds for full SIOD with 100 clones), with the main computational bottleneck being the final fine optimization, which has to re-propagate the hypothesis orbit on every loss function evaluation. Due to the processing time, we only ran SIOD for a limited subset of tracks in our dataset, and used synthetic SIOD for model training and baseline evaluation. This computation time could be reduced in the future using parallelization and tuning the hyperparameters of the fine optimization (such as using a larger tolerance to reduce the number of iterations).

3.2.3 Candidate Determination

Candidate determination creates a list of potential pairings between tracks and element sets. Pairings are considered “correct” if both the track and element set have the same satellite catalog number. Each track is paired with all element sets with epochs between 24 hours and one hour before the first observation of the track, with filters on the differences in semi-major axis, eccentricity, inclination, and orbital angular momentum between the element set and the mean SIOD estimate for the track. The filter thresholds were chosen to retain all correct pairings while filtering out as many incorrect pairings as possible. We excluded argument of perigee, right ascension of the ascending node, and mean/true anomaly because these values can differ widely from the reported values while describing approximately the same orbit when the orbit is near-circular and equatorial. After running candidate determination we ended up with 33.6 million candidate pairs, of which 6.4 million (19.0%) were correct pairings. We used smaller subsets of these candidate pairs in downstream steps to reduce processing time.

¹<https://pypi.org/project/sgp4/>

3.2.4 Probabilistic Kinematic Correlation

With our filtered subset of candidate pairs, the next objective is to compute a statistical correlation between the orbits of the track and element set in each remaining pair. This correlation represents the probability that the UCT (track) belongs to its paired catalog object or prior UCT (element set).

This procedure can be divided into three main parts: (1) State and Covariance Propagation, (2) Track Correlation, and (3) Kinematic Feasibility Determination. The first step, Covariance Propagation, starts with the end result of the SIOD process, i.e., a mean and covariance matrix of the state vector. For each track, we generate a set of orbital clones drawn from the multivariate normal distribution defined by its mean and covariance matrix. These orbital clones act as a representative sample of orbits that define a PDF of the position and velocity of the target observed in the track.

Using SGP4 [20], we propagate the orbits of the orbital clones for each track backwards in time for 24 hours (the maximum time difference for candidate pairs, see Section 3.2.1). The result of this propagation is a time-series of PDFs that define the state vector and corresponding uncertainties of the track over the correlation timeframe. Similarly, we use the state vectors and uncertainties reported with the element sets to generate orbital clones, which we propagate forward in time by 24 hours. This results in a series of state vector PDFs at times that overlap the track PDFs. Fig. 3 shows a schematic representation of this process.

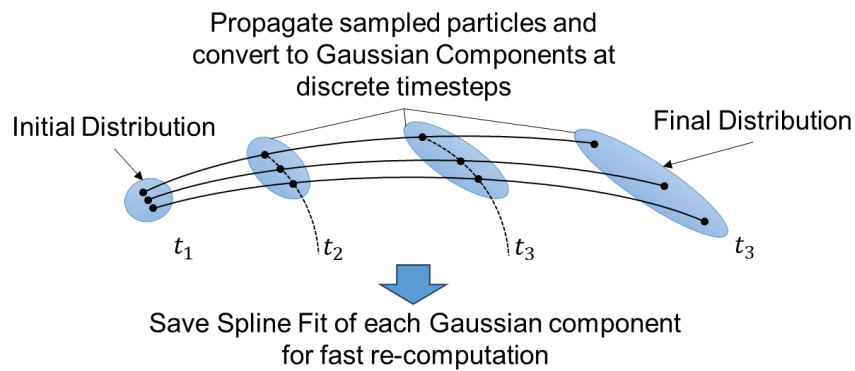


Fig. 3: Schematic of probabilistic orbit propagation.

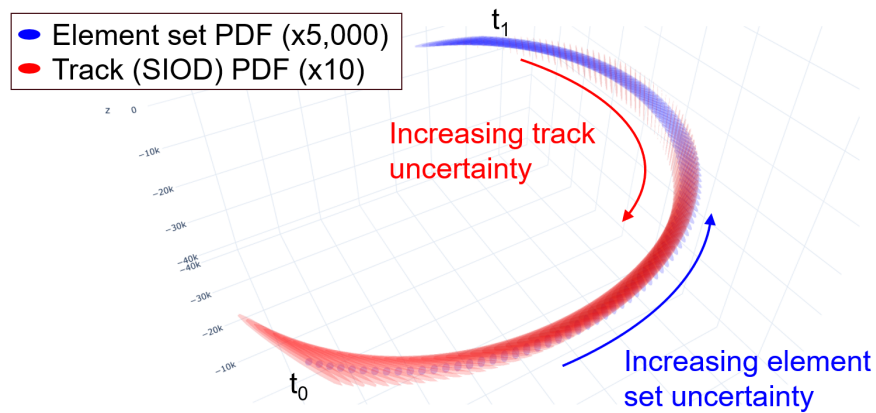


Fig. 4: Probability Density Function propagation of a catalog element set (blue) and a paired track orbit (red) determined with the SIOD step. The element set is propagated forward in time and its size has been exaggerated by a factor of 5,000. The track orbit is backwards propagated to matching times and has been enlarged by a factor of 10 for visual comparison. At t_0 , the element set has low uncertainty which grows as it is forward propagated and the track orbit has high uncertainty as it is propagated backwards in time from the time of observation t_1 .

In order to increase computational efficiency, we sample the PDFs of the orbital clones at discrete timestep intervals. This ensures that the track PDFs and element set PDFs align. Fig. 4 shows an example candidate pair with overlapping propagated PDFs. Additionally, we fit parametric splines through the mean states and covariance matrices and we then save those splines to a local database. These splines provide an accurate representation of the PDFs as a function of time and allow for fast interpolation between timesteps. This facilitates the large number of comparisons required to evaluate candidate pairs downstream in the pipeline. The spline database reduces the overall computational cost, since orbital clones for a given track or element set only need to be propagated once, even if they are present in multiple candidate pairs.

Once PDFs of each filtered candidate pair have been generated at a series of discrete intervals, they can be evaluated in the Track Correlation step. In Track Correlation, the relative position and velocity of the two PDFs are compared at each overlapping time. We do this by computing the Kullback-Leibler (KL) divergence, or relative entropy, which is a measure of how different two probability distributions are from one another [22]. It is defined as

$$D_{\text{KL}}(P \parallel Q) = \sum_{x \in \chi} P(x) \log \left(\frac{P(x)}{Q(x)} \right), \quad (1)$$

where $D_{\text{KL}}(P \parallel Q)$ is the relative entropy from Q to P , and Q and P are probability distributions dependant on variable(s) x within the sample space χ .

To determine the kinematic feasibility metrics (step 3) of each filtered candidate pair, we compute the minimum of the KL divergence over each overlapping time. This minimum corresponds to the time where the track and element set within the pair are most closely matched. The value of the KL divergence at this minimum, along with the position and velocity differences are passed to the confidence determination model to determine the likelihood that the given track belongs to the object.

3.2.5 Statistical Object Identification

The Statistical Object Identification step compares the non-kinematic observable information associated with a UCT—in this case the visual magnitude and solar phase angle from each observation—with known prior distributions for objects to produce an “in-family” score for that object. The in-family score is a numerical value that represents object conformity ranging from 0 to 1. A threshold can be set (e.g., 0.5) to classify objects as in-family or outliers for a given object distribution. Fig. 5 is an example of the in-family score as a function of the range normalized magnitude for a single RSO object type.

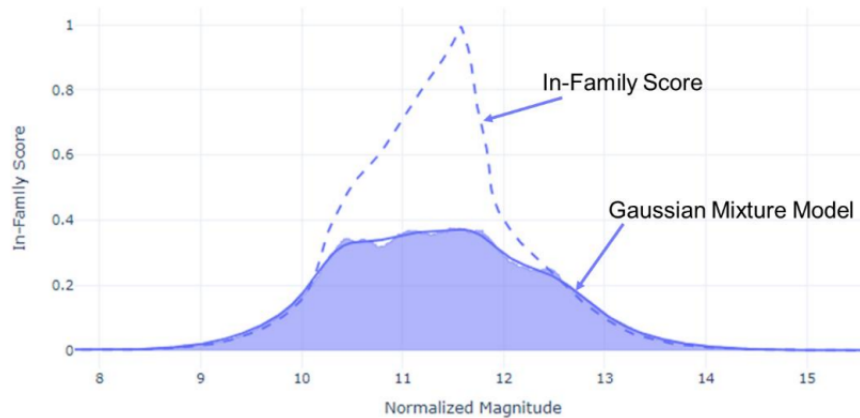


Fig. 5: An example Gaussian Mixture Model probability density function and the relation to in-family score.

The magnitude values from the historical observations (EXO and PPEC) were stored for each object and normalized using a diffuse sphere model at a reference range R_0 and phase angle of 0° . Visual magnitude can be estimated using

the following equation from [23]:

$$m_v = m_{\text{sun}} - 2.5 \log_{10} \left[\frac{\rho A_{\text{opt}}}{R^2} F(\phi) \right], \quad (2)$$

where ρ is the reflectivity, A_{opt} is the surface area, R is the range, ϕ is the solar phase angle, and $F(\phi)$ is the fractional diffuse reflection (also known as the phase function). For this normalization, we use the fractional diffuse reflection for a diffuse sphere [24]:

$$F(\phi) = \frac{2}{3\pi^2} [(\pi - \phi) \cos(\phi) + \sin(\phi)]. \quad (3)$$

Since m_v , ϕ , and R are all known for correlated EO observations, the area-albedo product ρA_{opt} can be estimated by rearranging Eq. 1:

$$(\rho A)_{\text{est}} = \left(\frac{R^2}{F(\phi)} \right) \cdot 10^{0.4(m_{\text{sun}} - m_v)}. \quad (4)$$

We then use this quantity in Eq. 1 to get the *Normalized Magnitude* m_{norm} , which represents the magnitude the object would have if it were at reference range R_0 and reference phase ϕ_0 , according to the diffuse sphere model:

$$m_{\text{norm}} = m_{\text{sun}} - 2.5 \log_{10} \left[\frac{(\rho A)_{\text{est}}}{R_0^2} F(\phi_0) \right]. \quad (5)$$

Assuming $\phi_0 = 0$, the function $F(\phi_0)$ can be simplified as follows:

$$F(0) = \frac{2}{3\pi^2} [(\pi - 0) \cos(0) + \sin(0)] = \frac{2}{3\pi}. \quad (6)$$

Hence,

$$m_{\text{norm}} = m_{\text{sun}} - 2.5 \log_{10} \left[\frac{(\rho A)_{\text{est}}}{R_0^2} \cdot \frac{2}{3\pi} \right]. \quad (7)$$

After normalizing the data, we bin them into 10 degree solar phase angle ranges from 0 to 150 degrees. Then, we scale the data and fit a separate 1-D Gaussian Mixture Model (GMM) [25] to each bin when more than 50 samples are present. This limits variation in the magnitude caused by changes in the solar phase angle to further isolate individual object signatures. After fitting the GMMs, a post processing pipeline transforms the GMM probabilities into an in-family score for each model, facilitating standardized comparisons between models.

3.2.6 Confidence Determination

The final step in our UCT correlation pipeline is Confidence Determination. During this step, we synthesize all of the results of previous steps in order to generate a robust statistical representation of the probability that each track belongs to a given element set. To do this, we implement a supervised learning approach to predict the correctness of pairings using a binary random forest classifier [26], as depicted in Fig. 6.

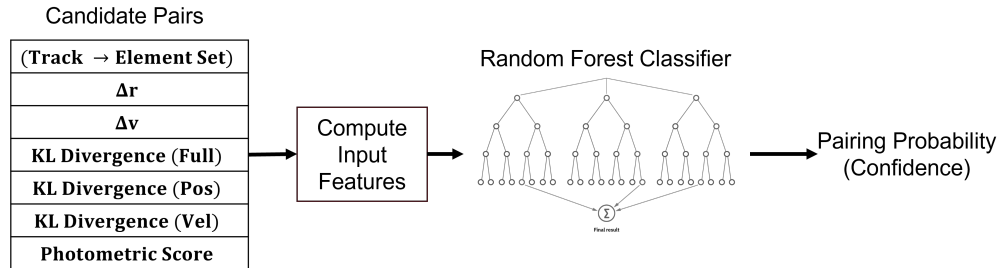


Fig. 6: Confidence determination architecture.

The inputs to the Confidence Determination algorithm are drawn from the outputs from Probabilistic Kinematic Correlation and Statistical Object Identification. From these inputs, the features given in Table 1 are calculated as inputs to

Table 1: Random Forest Classifier Input Features

Symbol	Description
Δv	The velocity difference between the track mean backward propagated state vector and element set mean forward propagated state vector.
Δr	The distance between the track mean backward propagated position and element set mean forward propagated position.
KL_{vel}	The KL divergence between 3-dimensional velocity probability distributions of the element set and the track.
KL_{pos}	The KL divergence between 3-dimensional positional probability distributions of the element set and the track.
KL_{full}	The KL divergence between the full 6-dimensional probability distributions of the element set and the track.
$IsMin\Delta r$	Boolean flag that is true when the candidate pair has the lowest position difference compared to any other candidates for the same track.
$IsMin\Delta v$	Boolean flag that is true when the candidate pair has the lowest velocity difference compared to any other candidates for the same track.
S_{phot}	The photometric in-family score for the track-catalog object pairing.
$IsMinKL_{vel}$	Boolean flag that is true when the candidate pair has the lowest velocity KL divergence compared to any other candidates for the same track.
$IsMinKL_{pos}$	Boolean flag that is true when the candidate pair has the lowest positional KL divergence compared to any other candidates for the same track.
$IsMinKL_{full}$	Boolean flag that is true when the candidate pair has the lowest full KL divergence compared to any other candidates for the same track.
$IsMinS_{phot}$	Boolean flag that is true when the candidate pair has the lowest photometric in-family score compared to any other candidates for the same track.
$IsMaxS_{phot}$	Boolean flag that is true when the candidate pair has the highest photometric in-family score compared to any other candidates for the same track.

the random forest classifier. All values are calculated at the time of minimum full KL divergence between the element set and the track. The relative importance for each of these features in the random forest classifier is shown in Fig. 7.

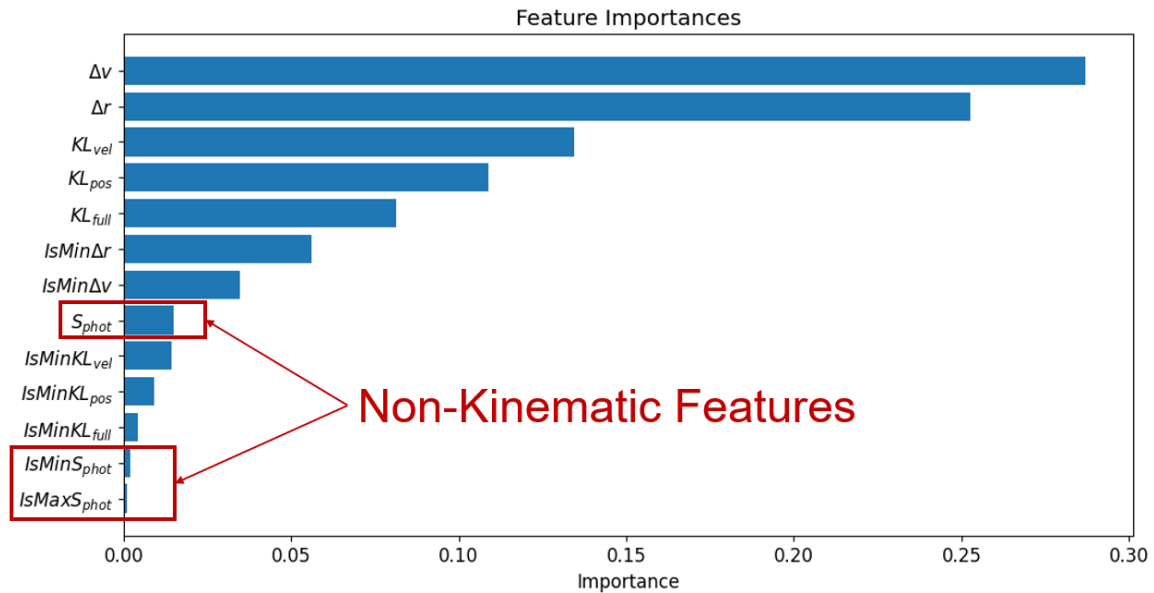


Fig. 7: Random Forest Feature Importance

After each candidate pair is classified, it receives a confidence score based on the distribution of classification scores between each element set paired with a given track. Confidence scores represent the likelihood that a candidate track-element set pair are the same object. Once these scores are determined, we compare the results from both our full hybrid (kinematic and non-kinematic) pipeline and our kinematic-only baseline version with the original data. Because we started with known correlated tracks, we can compare both models against the true distribution to effectively assess both relative and absolute performance.

4. RESULTS

In this section we test our hypothesis, that incorporating non-kinematic quantities in the track correlation process, in addition to kinematic quantities, will improve our ability to match UCTs to catalog objects (or previous UCTs). To do this, we test the same sample of our dataset using both our full hybrid pipeline and our kinematic-only baseline pipeline and compare the results of each with the known correlations within the data. A subset of our data have known maneuvers that occurred within the correlation timeframe. We highlight these examples throughout this section as a prime example of the effectiveness of our methods.

4.1 Stochastic Initial Orbit Determination

During initial testing, our SIOD method proved to be more accurate at estimating orbital parameters for single track observations than classical methods, such as those developed by Gauss [27] and Gooding [28]. We also found that this method was significantly faster at converging on an initial orbit than other sampling techniques, e.g., Markov Chain Monte Carlo simulations. In Fig. 8, we give an example of candidate best-fit orbits (red), generated by our SIOD step, in comparison with the true orbit (blue) of the known candidate.

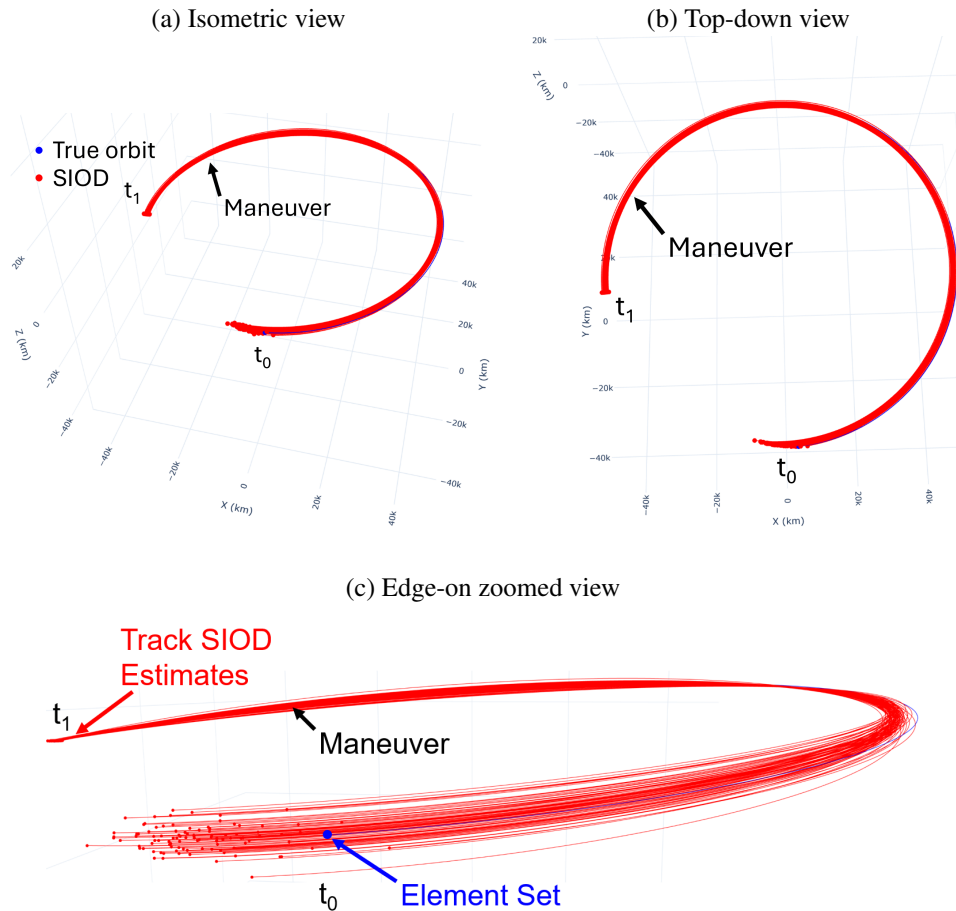


Fig. 8: True (blue) and recovered (red) orbits generated through the SIOD step for a track with a known maneuver.

This object is known to have carried out a maneuver within one day prior to the track observations. Despite the maneuver, our SIOD step adequately recovers the true orbit within its uncertainty distribution (spread of orbits at t_0).

We initially found that the SIOD algorithm would often converge on incorrect, high-eccentricity orbits (about half of the time). To address this we added an eccentricity constraint to the CAR of 0.01 to ensure the resulting orbits are very close to circular. In the future, we could evaluate SIOD at multiple eccentricity ranges and correlate the results of any solutions that converged to correlate to high-eccentricity objects.

4.2 Statistical Object Identification

The isolated performance of the Statistical Object Identification was evaluated using a stratified temporal split of observations from 200 distinct objects. The training set comprised 2.1 million observations and the test set was around 500,000 observations. It is noteworthy that the computational time required for the model to calculate in-family scores is negligible, so there is little to no added computational cost to incorporating this step into the hybrid pipeline.

The confusion matrix depicted in Fig. 9 was constructed using a hybrid threshold-maximum likelihood approach. For each object pair, we calculated the mean in-family score, employing a decision threshold of above 0.5 to determine the true label. This methodology can be framed as a combination of binary classification (for the true class) and multi-class maximum likelihood estimation. The diagonal elements of the confusion matrix represent the true positive rate (sensitivity) for each satellite number (satNo) model, conditioned on exceeding the 0.5 threshold. This approach was implemented to assess the average in-family prediction accuracy for each object model against the ground truth data.

The current model evaluates the normalized magnitude of test data only when the corresponding phase angle bin contains at least 50 observations. This threshold, while intended to only evaluate magnitude values observed within that phase angle bin, may exclude portions of the photometric behavior of the object.

It is important to note that this method assumes score calibration across different classes, which may not always hold in practice. The Random Forest model, used in the Confidence Determination step, would subsequently determine the specific confidence thresholds for final classification decisions, potentially mitigating any calibration discrepancies in the base Statistical Object Identification model.

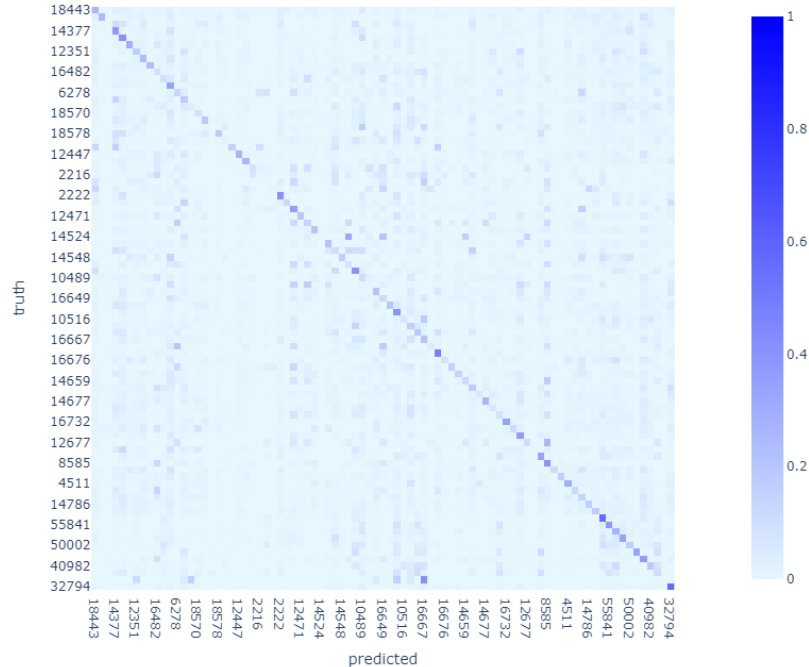


Fig. 9: Matrix representing the mean in-family scores for each true-predicted object pair. The diagonal elements represent the true positive rate (sensitivity) for each Object ID model. Off-diagonal elements indicate misclassifications or model confusion between different objects.

Fig. 10 shows the GMMs for object 58204 for subsets of solar phase angle bins. For each bin, the scaled density histogram of the training data is overlaid with the corresponding GMM probability density function and in-family score. The PDF and in-family score were computed using an array of 1,000 normalized magnitude values ranging from 5 to 25. To enhance visibility of the key features, the subplots focus on the distribution within the normalized magnitude range between 8 to 12. This visualization effectively demonstrates the model's ability to capture the underlying data distribution and its variation across different phase angles, providing insights into the photometric behavior of the object under varying observational conditions.

Object 58204 - Source EXO: Normalized Scores, PDF, and Data Histogram - Photometric Model

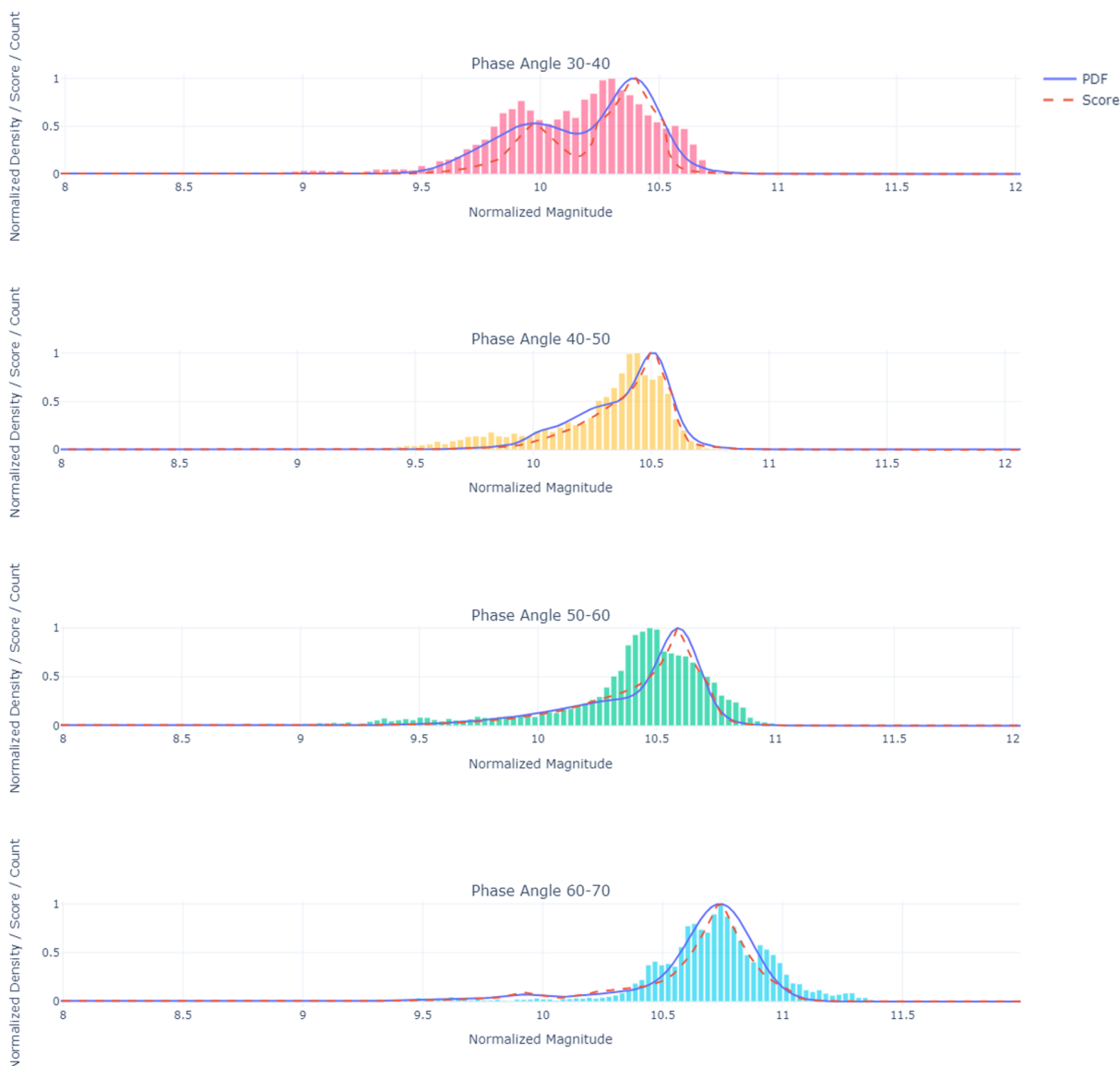


Fig. 10: A subset of the GMM models for object 58204 for each solar phase angle bin. The in-family score (red dashed) and PDF (blue solid) curves are shown for the normalized magnitude as well as the training data that was used.

4.3 Probabilistic Kinematic Correlation

Probabilistic Kinematic Correlation (PKC) compares the probability distributions of the forward-propagated element set and the backward-propagated SIOD estimate at intermediate times to find the time at which the full KL Divergence

between the two is minimized. At this time the KL divergence (full, position-only, and velocity), position error, and velocity error are calculated and saved for downstream processing. Figure 11 illustrates the full KL divergence over time and probability distributions from an example track with a reported maneuver. Fig. 12 shows the relationship between the probability of a given pairing being correct and the minimum KL divergence.

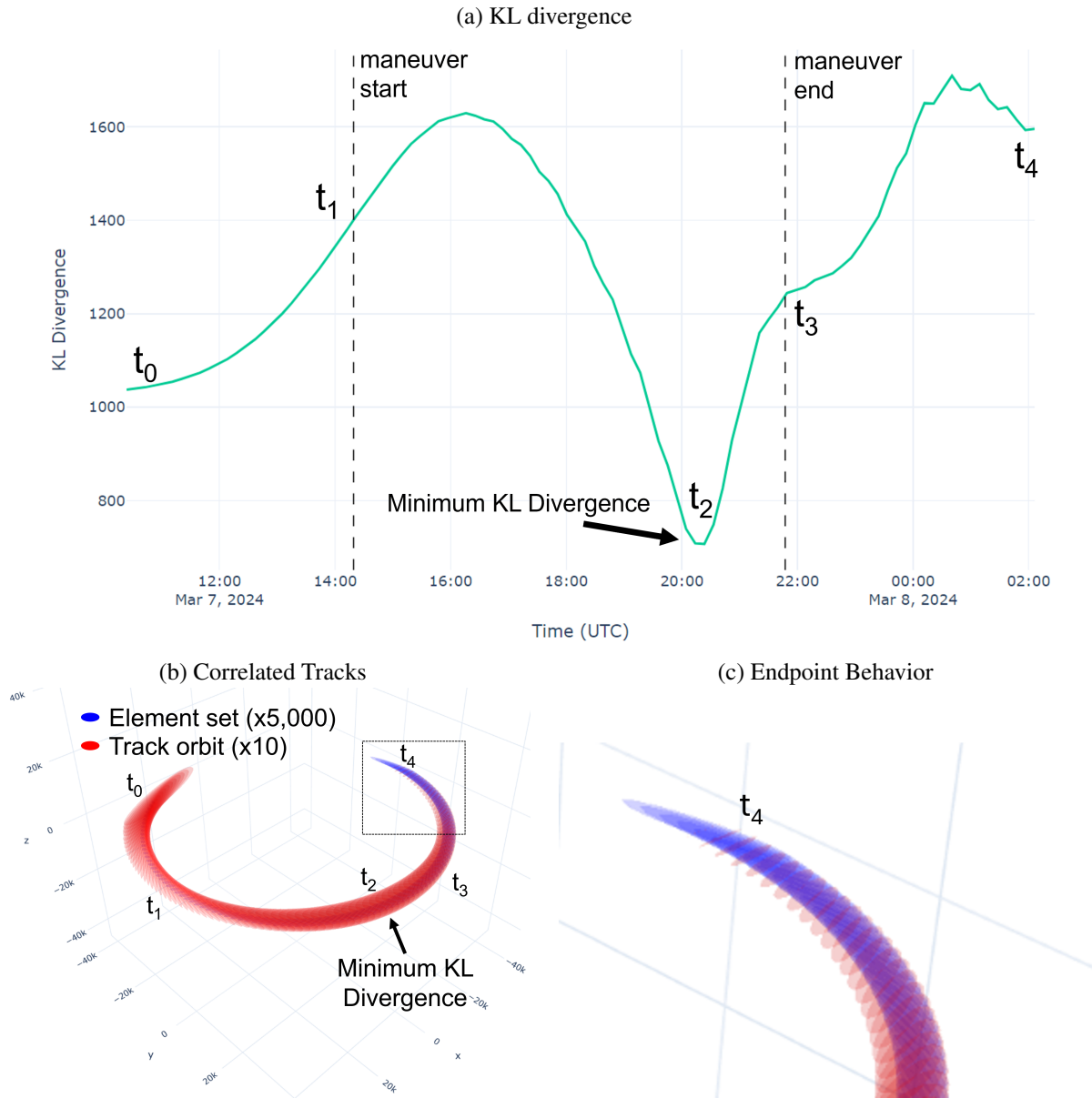


Fig. 11: Probabilistic Kinematic Correlation Visualization. (a) KL Divergence as a function of time for a example track. A known maneuver occurred during the time between the two vertical dashed lines (t_1 to t_3). The minimum, at t_2 , coincides with the best agreement between the PDFs of the track and element set. (b) Track (red) and element set (blue) propagated PDF uncertainty ellipsoids. Element set ellipsoids are propagated forward in time (starting from t_0) and track orbit ellipsoids are propagated backwards in time (starting from t_4). The black arrow indicates the location of the minimum KL divergence in (a). (c) An enlarged portion of the boxed region in (b). Here we see that track orbit uncertainty ellipsoids have low uncertainties and primarily in the cross track direction. Element set uncertainties are large at this later time and extend primarily along the orbit. Similar but reversed behavior can be seen at the initial time near t_0 in (b). Red track uncertainties have been enlarged by a factor of 10 and element set uncertainties have been inflated by a factor of 5,000 in order to facilitate visual comparison.

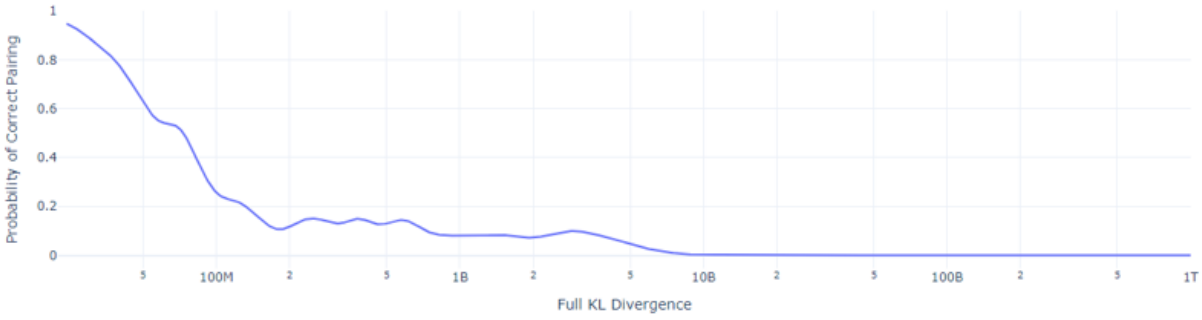


Fig. 12: Pairing Probability vs Minimum KL Divergence

4.4 Confidence Determination

We trained the confidence determination model using the results from synthetic SIOD described in Section 3.2.2. We evaluated this model using a baseline test dataset, which also uses synthetic SIOD, as well as an evaluation dataset generated using results from real SIOD with tracks where recent maneuvers were reported. All tracks in either of the test datasets were excluded from the training dataset. Two variants of confidence determination were trained: one with only features derived from kinematic information (the “Kinematic Model”), and the other also including non-kinematic features (the “Hybrid Model”). Table 2 shows the performance of both of these models on the evaluation datasets.

Table 2: Classification Reports for Confidence Determination

(a) Kinematic Model, synthetic SIOD results					(b) Hybrid Model, synthetic SIOD results				
	precision	recall	f1-score	support		precision	recall	f1-score	support
False	1.00	1.00	1.00	40801	False	1.00	1.00	1.00	40801
True	0.98	0.95	0.97	621	True	0.99	0.95	0.97	621
accuracy			1.00	41422	accuracy			1.00	41422
macro avg	0.99	0.97	0.98	41422	macro avg	0.99	0.97	0.98	41422
weighted avg	1.00	1.00	1.00	41422	weighted avg	1.00	1.00	1.00	41422

(c) Kinematic Model, real SIOD results					(d) Hybrid Model, real SIOD results				
	precision	recall	f1-score	support		precision	recall	f1-score	support
False	0.99	0.99	0.99	9264	False	1.00	0.99	0.99	9264
True	0.94	0.96	0.95	1714	True	0.95	0.97	0.96	1714
accuracy			0.98	10978	accuracy			0.99	10978
macro avg	0.97	0.97	0.97	10978	macro avg	0.97	0.98	0.98	10978
weighted avg	0.98	0.98	0.98	10978	weighted avg	0.99	0.99	0.99	10978

As seen in Table 2, the incorporation of the photometric in-family scores made a marginal improvement in the ability to correctly associate tracks to element sets. The limited improvement is likely due to a combination of the limited coverage of solar phase angles in the observation dataset, many objects in similar orbits displaying similar photometric signatures, and the high baseline performance of the purely kinematic model. Fig. 13 and Fig. 14 show the precision-recall and receiver operating characteristic (ROC) curves for both the kinematic and hybrid models, evaluated using the real SIOD dataset.

Precision-Recall Curve

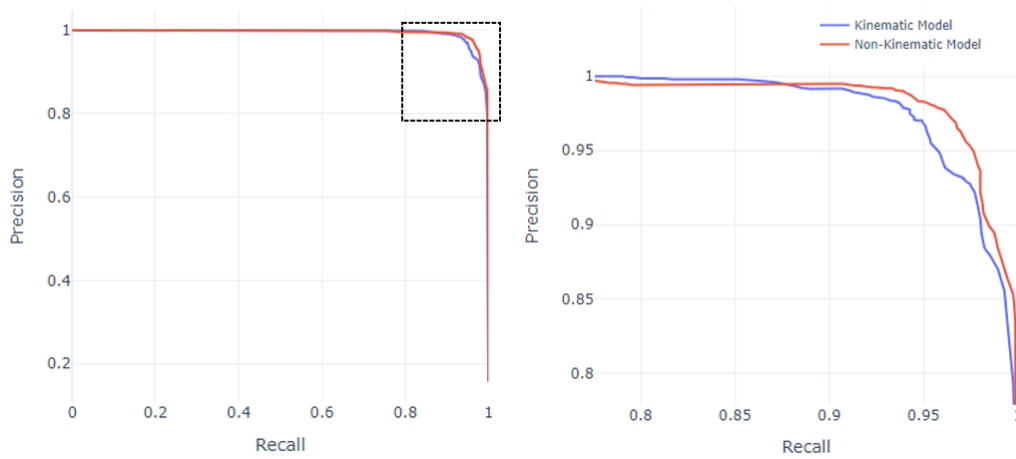


Fig. 13: Confidence Determination precision-recall curve (evaluated using real SIOD results)

ROC Curve

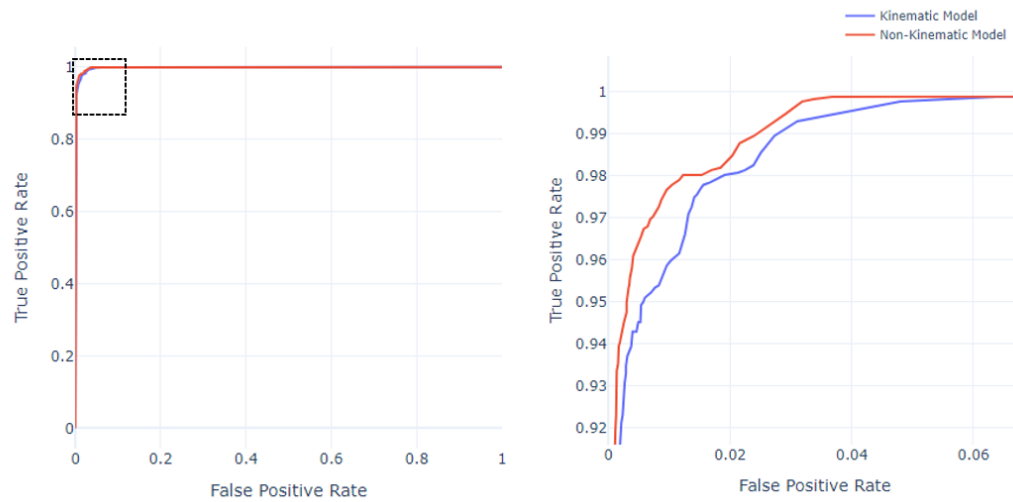


Fig. 14: Confidence Determination ROC curve (evaluated using real SIOD results)

One important feature of Confidence Determination is to provide a reliable mapping from the reported confidence to the actual probability that the track belongs to a given object. Fig. 15 and Fig. 16 show the relationship between the reported probability (confidence) of the model and the actual probability of a track belonging to a particular object.



Fig. 15: Pairing probability vs confidence (evaluated using synthetic SIOD results)

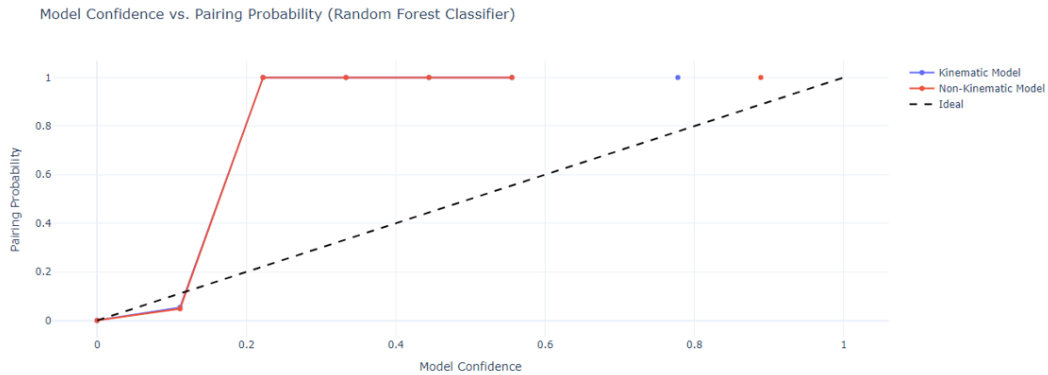


Fig. 16: Pairing probability vs confidence (evaluated using real SIOD results)

As seen in Fig. 16, while the model remains highly predictive for real SIOD data, it loses its clean linear mapping from confidence to true probability. This is most likely due to the fact that the evaluation dataset (with real SIOD) is drawn from a different distribution from the training dataset (which uses synthetic SIOD). One potential way to improve this would be to use results from real SIOD for both training and evaluation.

5. CONCLUSION AND FUTURE WORK

In this study, we developed and evaluated a UCT resolution workflow that integrates both kinematic and non-kinematic data to improve the correlation of tracks to cataloged objects. The Stochastic Initial Orbit Determination (SIOD) step proved effective in estimating orbital parameters from single-track observations, even in the presence of maneuvers. While the incorporation of photometric information through Statistical Object Identification showed promise, its impact was limited due to factors such as incomplete phase angle coverage and similarities in photometric signatures among objects in similar orbits. Our Confidence Determination model achieved high accuracy in correctly associating tracks to element sets, with the hybrid model slightly outperforming the kinematic-only model. While this approach shows high performance on our dataset, this is not necessarily representative of real-life performance for UCT resolution. This is because our truth labels were created from observations that have been already correlated.

This research presents several avenues for future work, particularly in additional training and evaluation, expanded use cases, and further exploration into object identification using non-kinematic data. Firstly, the performance on post-maneuver tracks and uncorrelated tracks should be more thoroughly evaluated. This can be done by simulating maneuvers of various sizes and gauging the effect on correlation performance, and by identifying tracks that were initially uncorrelated, but were identified later when more data was available. Secondly, the processing pipeline should be expanded to work with multiple eccentricity ranges. We found that SIOD often converged on incorrect, high-

eccentricity orbits if we did not provide a tight constraint on eccentricity. Because of this, the workflow is restricted to correlating UCTs to low-eccentricity orbits. In the future SIOD can be evaluated with multiple eccentricity ranges and all results that converged can be utilized in the downstream correlation algorithms. Finally, additional methods for object identification using non-kinematic information could be explored. This includes exploration of additional phase functions for different shapes and combinations of specular and diffuse reflections for computation of normalized magnitude, different statistical and/or machine learning models for comparing tracks to objects, as well as the use of additional phenomenologies such as radar cross section, passive radio frequency signature, and spectral information.

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